

Michealmas Week 1 & 2

lw703

1 Completed Task

I have read the paper "Training Neural Samplers with Reverse Diffusive KL Divergence". With this paper, I got to understand what is neural sampler.

Formally, many neural samplers learn a parameterized transformation $f_\theta : \mathcal{Z} \rightarrow \mathcal{X}$, where $\mathcal{Z}$ is a latent space endowed with a simple, easy-to-sample prior distribution $p_Z(\mathbf{z})$ (e.g., a standard multivariate Gaussian $\mathcal{N}(0, \mathbf{I})$). The neural network f_θ , with parameters θ , maps a latent variable $\mathbf{z} \sim p_Z(\mathbf{z})$ to a sample $\mathbf{x} = f_\theta(\mathbf{z})$ in the target data space $\mathcal{X}$. The parameters θ are optimized such that the distribution of generated samples $p_\theta(\mathbf{x})$ closely matches the target distribution $p_d(\mathbf{x})$.

In DiKL paper, the loss is defined as:

$$\text{DiKL}(p_\theta \| p_d) = \sum_{i=1}^{N_k} w_i \cdot \text{KL}((p_\theta * k_i) \| (p_d * k_i)), \quad (1)$$

where $\{k_i\}_{i=1}^{N_k}$ is a set of Gaussian convolution kernels (typically with varying bandwidths), w_i are positive weights, and $*$ denotes the convolution operation.

The gradient is derived as:

$$\begin{aligned} \nabla_\theta \text{KL}(p_{\theta,i} \| p_{d,i}) &= \nabla_\theta \mathbb{E}_{\mathbf{x}_i \sim p_{\theta,i}} \left[\log \frac{p_{\theta,i}(\mathbf{x}_i)}{p_{d,i}(\mathbf{x}_i)} \right] \\ &= \mathbb{E}_{\mathbf{z}, \epsilon_i} \left[(\nabla_{\mathbf{x}_i} \log p_{\theta,i}(\mathbf{x}_i) - \nabla_{\mathbf{x}_i} \log p_{d,i}(\mathbf{x}_i)) \frac{\partial \mathbf{x}_i}{\partial \theta} \right], \end{aligned} \quad (2)$$

where $\mathbf{x}_i$ in the score terms is the sample generated via $g_\theta(\mathbf{z}, \epsilon_i)$, and $\frac{\partial \mathbf{x}_i}{\partial \theta}$ is the Jacobian of the sampling path with respect to θ .

Meanwhile, I found an alternative to derive the gradient of the loss:

$$\begin{aligned} \nabla_\theta D_{KL}(p_\theta \| p_d) &= \int p_\theta(x_t) (\nabla_{x_t} \log p_\theta(x_t) - \nabla_{x_t} \log p_d(x_t)) \frac{\partial x_t}{\partial \theta} dx_t \\ &= \int \int \nabla_{x_t} \log k(x_t | x) p_\theta(x | x_t) dx p_\theta(x_t) \frac{\partial x_t}{\partial \theta} dx_t \\ &\quad - \int \int \nabla_{x_t} \log p_d(x_t) \frac{\partial x_t}{\partial \theta} p_\theta(x_t, x) dx_t dx \\ &= \int \int (\nabla_{x_t} \log k(x_t | x) - \nabla_{x_t} \log p_d(x_t)) \frac{\partial x_t}{\partial \theta} p_\theta(x_t, x) dx_t dx \\ &= \int \int \left(\frac{\alpha x - x_t}{\beta^2} - \nabla_{x_t} \log p_d(x_t) \right) \frac{\partial x_t}{\partial \theta} p_\theta(x_t, x) dx_t dx \end{aligned}$$

$$\nabla_\theta D_{KL}(p_\theta \| p_d) = \int \left(\frac{\alpha E_{p(x|x_t)}[x] - x_t}{\beta^2} - \nabla_{x_t} \log p_d(x_t) \right) \frac{\partial x_t}{\partial \theta} p_\theta(x_t) dx_t$$